

Unit 1, Lesson 1: Statistics and Interpreting Data

Benchmarks

6.DP.1.1 Recognize and formulate a statistical question that would generate numerical data.

6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.

Learning Targets

- ☐ I can identify statistical questions and classify types of data.
- ☐ I can interpret information from a data set or a graphical display.

1. Alejandro, a mushroom-kit entrepreneur, says that when it comes to growing his business, "The most important thing is ... adapting opportunities that come to you and still holding the same amount of passion."
 - a. What is something you are passionate about?
 - b. How has a new interest or hobby surprised you?

1.1.2 Refresher: Defining a Statistical Question



A **statistical question** is a question that can be answered by collecting data. Often, there will be variability in the data.

Examples and non-examples of statistical questions are shown.

Statistical Questions	NOT Statistical Questions
• How many siblings do the students in my class have? • How tall are the players on the girls’ basketball team? • What is the favorite song of the 6th graders at my school?	• What time does school start? • Who is the President of the United States? • What is Avery’s favorite color?

1. What do you notice about the two types of questions?
2. What do you think “variability in the data” means in the definition of “statistical question?”
3. Explain why the question, “How many times do the students at my school go swimming this summer?” is a statistical question.

1.1.3 Try It: Identifying Statistical Questions

1. Sort the following questions into the appropriate column by writing the letters of the questions in the table provided.
- a.

How many miles do the students in my class live from the school campus?
- b.

How many counties are there in Florida?
- c.

Who is the governor of Florida?
- d.

How many siblings do my 7th grade teachers have?
- e.

How many hours do adults in Florida spend outdoors on a weekend?
- f.

How many days are in December?

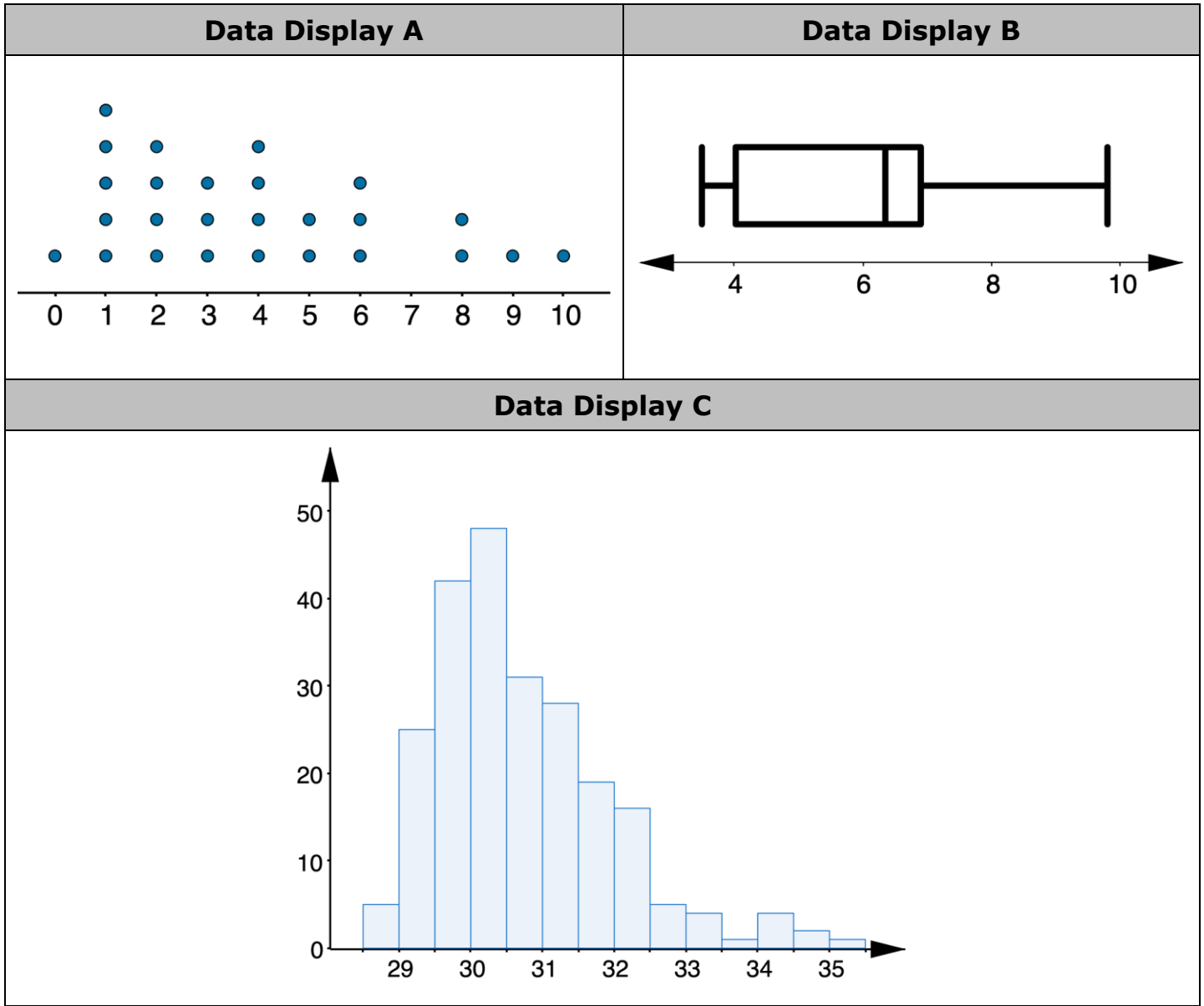
Statistical Questions	NOT Statistical Questions

1.1.4 Refresher: Data Displays

Numerical data can be displayed graphically in a variety of ways. Three graphs that are commonly used are histograms, line plots, and box plots.

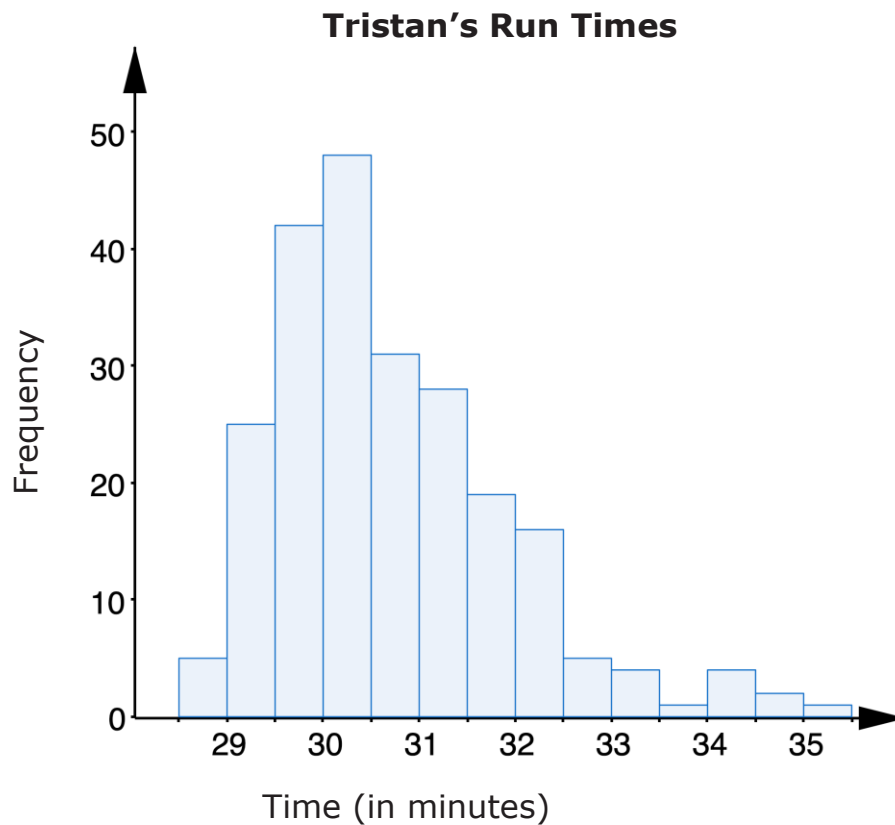
1. Three data displays are shown below. Match each type of data display with the examples shown by writing the letter on the line.

Histogram _____ Line Plot _____ Box Plot _____



1.1.5 Guided Instruction: Interpreting Histograms and Line Plots

1. Tristan has been tracking the time, in minutes, it takes him to run 4 miles for the last 10 years. A histogram of Tristan's run times is shown.



- a. The shape of run times can be best described

as

- ☐ symmetrical.
- ☐ right skewed.
- ☐ left skewed.

- b. Tristan's typical run time was about _____

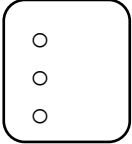
minutes. Most of his run times were between

_____ to _____ minutes. Not many of his runs

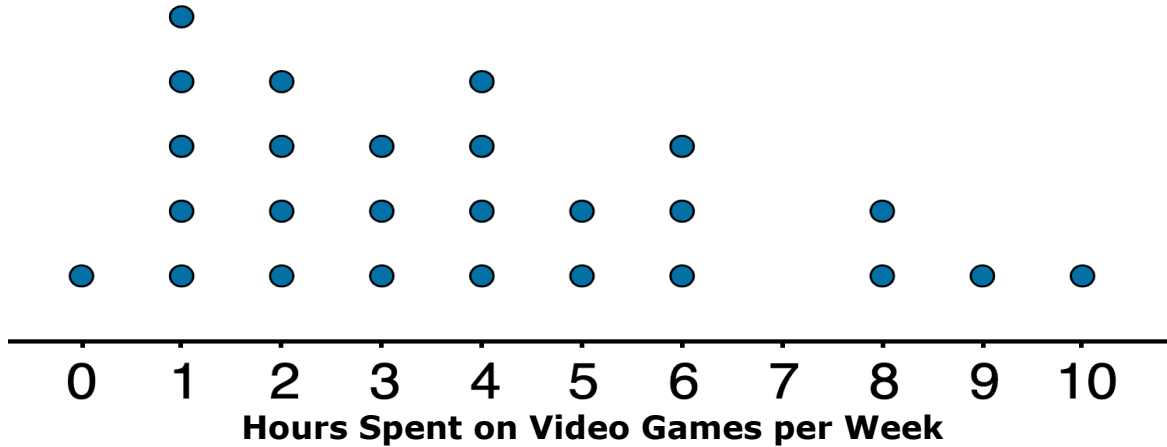
were

- ☐ shorter
- ☐ longer

than 32.5 minutes.

- c. If the data value  was added to the histogram, it would be considered an outlier.

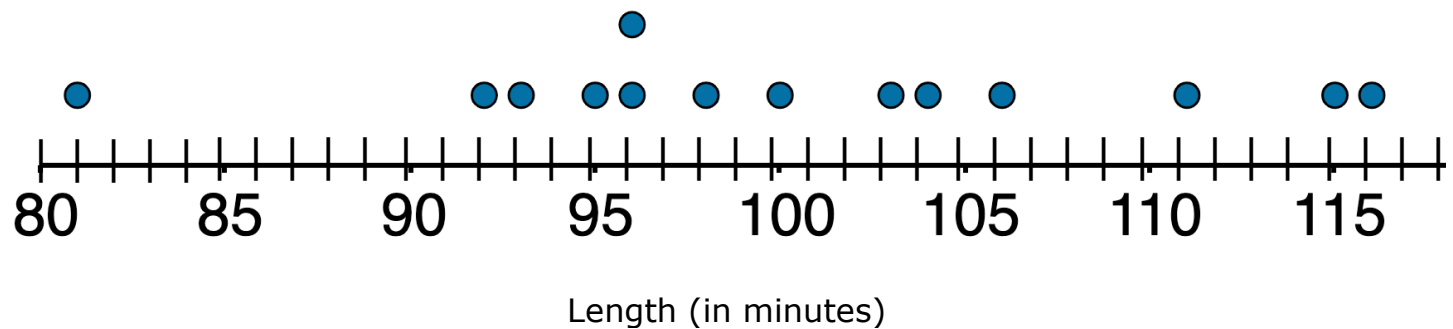
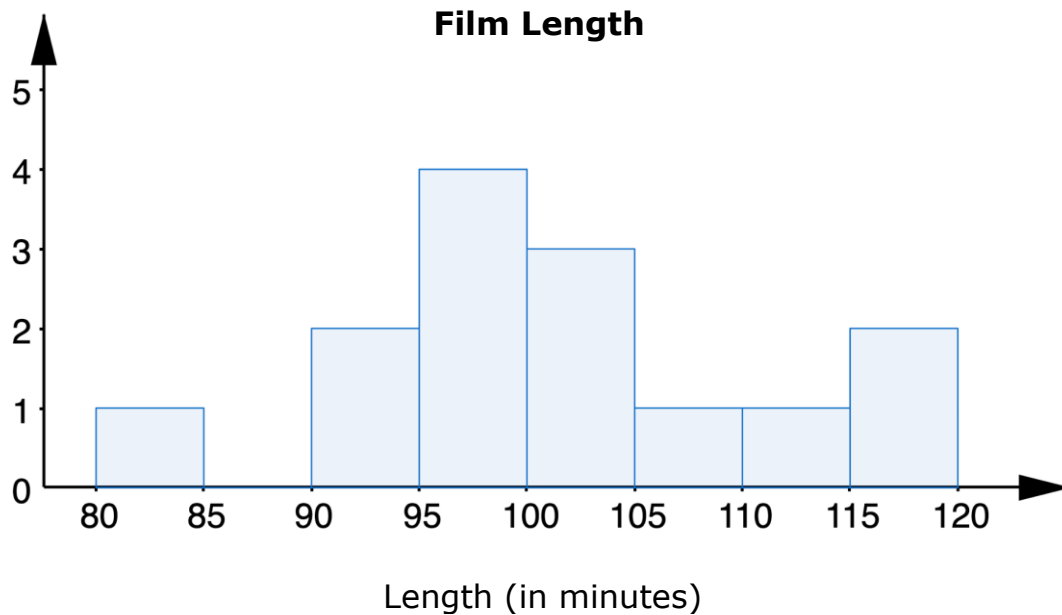
2. The line plot shows the number of hours per week that students reported playing video games.



- a. How many students were surveyed?
- b. How many students reported spending 1 hour on playing video games each week?
- c. What percentage of the students reported spending 4 hours or fewer on video games each week?
- d. What is the range of time spent playing video games in this group?

1.1.6 Collaboration: Which Graph?

1. A movie critic recorded the length, in minutes, of a sample of Pixar films and made a histogram and a line plot of the data.



- a. Which graph could tell you how many film lengths were recorded?
- b. How many film lengths are recorded?

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- c. How many films lasted at least 95 minutes but fewer than 100 minutes, according to the histogram?
 - d. How many films lasted at least 95 minutes but fewer than 100 minutes, according to the line plot?
 - e. What is the range of the film lengths?
 - f. Work with your partner to explain which graph was used to determine the range.
 - g. Are there any outliers? If there are, name them.

1.1.7 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of statistics and interpreting data.

